DATA SCIENCE & ARTIFICIAL INTELLIGENCE



Calculus and Optimization

Integration

Lecture No. 02



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Recap of previous lecture



Topic MAXIMA-MINIMA

Topics to be Covered



Topic INTEGRATION

INTEGRATION

① $y = \sin(e^{x^2})$ then $\frac{dy}{dx} = ? = \cos(e^{x^2}) \cdot e^{x^2} \cdot (2x)$

& $\int \sin(e^{x^2})\,dx = ?$ = Not possible to integrate

Int is tough because chain Rule DNE

② Area $= \iint (1)\,dx\,dy = \int y\,dx = \int_{x=a}^{b} f(x)\,dx = ?$

Volume $= \iiint (1)\,dx\,dy\,dz = \iint z\,dx\,dy = \iint f(x,y)\,dx\,dy = ?$

Mass $= \iiint \rho \cdot dx\,dy\,dz$

(3) Linear Algebra, Prob & Stats, Calculus ↘ Diff / Integration

→ Single Int / Multiple Int, $\int_0^\infty \frac{\sin x}{x}\,dx = ?$ (Laplace Transf), $\int_0^\infty e^{-x^2}\,dx = ?$ (β-γ func^n), $\int$(complex func^n)$\,dz = ?$ (C-I Th / C-R Th)

$\int$(vector)$\,d\vec{r}$ (G-Div / G-Th / S Th), C.R.V., $P(a<x<b) = \int_a^b f(x)\,dx$ (Prob & Stats), $\frac{d^2y}{dx^2} + y = 0$ (DEq) → (twice Integrate)

$\int_1^5 (?)\,dx = ?$ → N. Tech (T Rule / Simpson's)

For CS & DA :-
1. L. Algebra
2. Calculus
3. Prob & Stats

INDEFINITE INT → Collection of all the Antiderivatives is called Indefinite Int.

if $\frac{d}{dx} f(x) = \phi(x)$ then $\boxed{\int \phi(x)\,dx = f(x) + C}$

where $\phi(x)$ = Derivative of $f(x)$ & $f(x)$ = Antiderivative (Primitive) of $\phi(x)$

eg: $\frac{d}{dx}(x^3) = 3x^2$

$\frac{d}{dx}(x^3+5) = 3x^2$

$\frac{d}{dx}(x^3-7) = 3x^2$

$\frac{d}{dx}(x^3+10) = 3x^2$

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$\Rightarrow \int (3x^2)\,dx = x^3 + C$

Methods of solving Indefinite Integration →

(M-I) Using Standard Results

(M-II) Using substitution Method (M. Imp)

(M-III) use Integration By parts

$$\int uv\,dx = u\int v\,dx - \int\left\{\frac{du}{dx}\int v\,dx\right\}dx + C$$

(M-IV) Using Partial fraction

Standard Results :-

(1) Power formula : $\int x^a dx = \frac{x^{a+1}}{a+1}, a \neq -1$

(2) $\int \frac{1}{x} dx = \log x + C$

(3) $\int a^x dx = \frac{a^x}{\log_e a} + C$

(4) $\int e^x dx = e^x + C$

(5) $\int \sin x dx = -\cos x + C$

(6) $\int \cos x dx = \sin x + C$

(7) $\int \tan x dx = \log \sec x + C$

(8) $\int \cot x dx = \log \sin x + C$

(9) $\int \sec x dx = \log(\sec x + \tan x) + C$

(10) $\int (\operatorname{cosec} x) dx = \log(\operatorname{cosec} x - \cot x) + C$

(11) $\int \sec^2 x dx = \tan x + C$

(12) $\int \operatorname{cosec}^2 x dx = -\cot x + C$

(13) $\int e^{ax} \sin bx dx = \frac{e^{ax}}{a^2+b^2}(a \sin bx - b \cos bx)$

(14) $\int e^{ax} \cos bx dx = \frac{e^{ax}}{a^2+b^2}(a \cos bx + b \sin bx)$

(15) $\int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right)+C$

(16) $\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\log\left(\frac{x-a}{x+a}\right)+C$

(17) $\int \frac{dx}{a^2-x^2} = \frac{1}{2a}\log\left(\frac{a+x}{a-x}\right)+C$

(18) $\int \frac{dx}{\sqrt{x^2+a^2}} = \log\left(x+\sqrt{x^2+a^2}\right)+C$

(19) $\int \frac{dx}{\sqrt{x^2-a^2}} = \log\left(x+\sqrt{x^2-a^2}\right)+C$

(20) $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right)+C$

(21) $\int \sqrt{x^2+a^2}\,dx = \frac{x}{2}\sqrt{x^2+a^2} + \frac{a^2}{2}\log\left(x+\sqrt{x^2+a^2}\right)+C$

(22) $\int \sqrt{x^2-a^2}\,dx = \frac{x}{2}\sqrt{x^2-a^2} - \frac{a^2}{2}\log\left(x+\sqrt{x^2-a^2}\right)+C$

(23) $\int \sqrt{a^2-x^2}\,dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\left(\frac{x}{a}\right)+C$

(24) $\int \sec x \tan x\,dx = \sec x + C$, (25) $\int (\operatorname{cosec} x \cot x)\,dx = -\operatorname{cosec} x + C$

(26) $\int \frac{f'(x)}{f(x)}dx = \log f(x) + C$,

eg $\int \frac{dx}{x} = ? = \log(x) + C$

eg $\int \tan x\,dx = ? = \int \left(\frac{\sin x}{\cos x}\right)dx$

$= -\int \frac{(-\sin x)}{\cos x}dx$

$= -\log(\cos x) + C$

$= \log(\sec x) + C$

(27) $\int e^x \{f(x) + f'(x)\}dx = e^x f(x) + C$

eg $\int e^x(\sin x + \cos x)dx = ? = e^x \sin x + C$

eg: $\int [\sin(\log x) + \cos(\log x)]dx = ?$ Ans: $x\sin(\log x) + C$

Put $\log_e x = t \Rightarrow x = e^t$

$dx = e^t \cdot dt$

$= \int (\sin t + \cos t)\, e^t dt$

$= \int e^t(\sin t + \cos t)dt = e^t \sin t + C$

$= x \sin(\log_e x) + C$

(*) $\boxed{\int x^a\,dx = \frac{x^{a+1}}{a+1}}$, $a \neq -1$; eg $\int k\,dx = ? = k\int x^0\,dx = k\left(\frac{x^{0+1}}{0+1}\right) = kx + C$

eg $\int (x^2)\,dx = ? = \frac{x^{2+1}}{2+1} = \frac{x^3}{3}$

eg $\int \left(\frac{1}{x^2}\right)dx = ? = \int x^{-2}\,dx = \frac{x^{-2+1}}{-2+1} = \frac{-1}{x}$

eg $\int (\sqrt{x})\,dx = ? = \int x^{\frac{1}{2}}\,dx = \frac{x^{\frac{1}{2}+1}}{\frac{1}{2}+1} = \frac{2}{3}x^{3/2}$

eg $\int \left(\frac{1}{\sqrt{x}}\right)dx = ? = \int x^{-\frac{1}{2}}\,dx = \frac{x^{-\frac{1}{2}+1}}{-\frac{1}{2}+1} = 2\sqrt{x}$

eg $\int (x)\,dx = ? = \frac{x^2}{2}$

eg $\int \left(\frac{1}{x}\right)dx = ? = \log_e x$

Qs $\int\left(\sqrt{x}-\frac{1}{\sqrt{x}}\right)^2 dx = ? = \int\left(x+\frac{1}{x}-2\right)dx = \frac{x^2}{2}+\log_e x - 2x + C$

Qs $I = \int\left(\frac{x}{3}+\frac{3}{x}+x^3+3^x\right)dx = ?$

$= \frac{1}{3}\left(\frac{x^2}{2}\right)+3\left(\log_e x\right)+\left(\frac{x^4}{4}\right)+\left(\frac{3^x}{\log_e 3}\right)$

$= \frac{x^2}{6}+3\ln x+\frac{x^4}{4}+\frac{3^x}{\log_e 3}$

Qs $I = \int\left(e^{x\log_e a}+e^{a\log_e x}+e^{a\log_e a}\right)dx = ?$

$= \int\left(e^{\log_e a^x}+e^{\log_e x^a}+e^{\log_e a^a}\right)dx$

$= \int\left(a^x+x^a+a^a\right)dx$

$= \frac{a^x}{\log_e a}+\frac{x^{a+1}}{a+1}+a^a(x)+C$

Q. $\int e^{3\log x}(x^4+1)^{-1}\,dx = ?$

$$= \int \frac{e^{\log_e(x^3)}}{x^4+1}\,dx$$

$$= \int \left(\frac{x^3}{x^4+1}\right)dx$$

$$= \frac{1}{4}\int \left(\frac{4x^3}{x^4+1}\right)dx$$

$$= \frac{1}{4}\log\left(x^4+1\right) + C$$

Q. $I = \int \frac{dx}{\sqrt{1-e^{2x}}} = ?$

$$= \int \frac{dx}{e^x\sqrt{e^{-2x}-1}} = \int \frac{e^{-x}}{\sqrt{e^{-2x}-1}}\,dx$$

Put $e^{-x} = t \Rightarrow -e^{-x}\,dx = dt$

$$I = \int \frac{-dt}{\sqrt{t^2-1}} = -\int \frac{1}{\sqrt{t^2-1}}\,dt$$

$$= -\log\left\{t+\sqrt{t^2-1}\right\}$$

$$= -\log\left\{e^{-x}+\sqrt{e^{-2x}-1}\right\}$$ Ans

(25) Integration by parts $\Rightarrow \int \underset{\text{I}}{u} \cdot \underset{\text{II}}{v}\,dx = u\int v\,dx - \int \left\{\frac{d}{dx}(u)\cdot\int v\,dx\right\}dx + c$



Inverse Trig func^n

Algebraic func^n

Exp. func^n

Shortcut - e

The letter which is coming 1st in the word ILATE should be assumed as 1st func^n 'u'

Q: $I=\int(x\log x)\,dx=?$

$=\int(\log x)\cdot x\,dx$

$u=\log x \quad v=\frac{x^2}{2}$

$=\log x\left(\frac{x^2}{2}\right)-\int\left\{\left(\frac{1}{x}\right)\left(\frac{x^2}{2}\right)\right\}dx+C$

$=\frac{x^2}{2}(\log x)-\frac{1}{2}\int x\,dx+C$

$=\frac{x^2}{2}(\log x)-\frac{x^2}{4}+C$

Q: (CE-2025) $\int(\log x)\,dx=?=\int(\log x)\cdot 1\,dx$

$u=\log x \quad v=x$

$=\log x(x)-\int\left\{\left(\frac{1}{x}\right)(x)\right\}dx+C$

$\int(\log x)dx=x(\log x)-x+C$ Learn

Q: $I=\int(\log x)^2\,dx=?$

$=\int(\log x)^2\cdot 1\,dx$

$=(\log x)^2(x)-\int\left\{2\log x\left(\frac{1}{x}\right)\cdot(x)\right\}dx$

$=x(\log x)^2-2\int\log x\,dx+C$

$=x(\log x)^2-2[x\log x-x]+C$

Q $I=\int \frac{1-\cos x}{\cos x(1+\cos x)}dx=?$ Ans: $\log(\sec x+\tan x)-2\tan\left(\frac{x}{2}\right)+C$

& P.f: $A=1,\ B=-2$

Sol: $\frac{1-\cos x}{\cos x(1+\cos x)}=\frac{A}{\cos x}+\frac{B}{1+\cos x}=\frac{1}{\cos x}+\frac{-2}{1+\cos x}$

$\Rightarrow I=\int\left(\frac{1}{\cos x}+\frac{-2}{1+\cos x}\right)dx$

$=\int \sec x\,dx-2\int\frac{1}{2\cos^2\frac{x}{2}}dx$

$=\log(\sec x+\tan x)-\int \sec^2\left(\frac{x}{2}\right)dx$

Put $\frac{x}{2}=t\Rightarrow dx=2dt$

$I=\log(\sec x+\tan x)-\int \sec^2 t\,(2dt)$

$=\log(\sec x+\tan x)-2\tan t.$

$=\log(\sec x+\tan x)-2\tan\left(\frac{x}{2}\right)$

Note: $\int x^3 dx = \frac{x^4}{4} \Rightarrow \int (2x+5)^3 dx = ? = \int t^3 \frac{dt}{2} = \frac{1}{2}\left(\frac{t^4}{4}\right) = \frac{(2x+5)^4}{8}$

$\int \sin x = -\cos x \Rightarrow \int \sin(5x) dx = ? = \frac{-\cos(5x)}{5}$ Ans

Put $2x+5 = t \Rightarrow dx = \frac{dt}{2}$

DEFINITE INTEGRAL

Fundamental Theorem of Integral Calculus →

$$\int_a^b f(x)\,dx = \Big[\phi(x)+c\Big]_{x=a}^{x=b}$$

$$= (\phi(b)+c) - (\phi(a)+c)$$

$$\boxed{\int_a^b f(x)\,dx = \phi(b) - \phi(a)}$$

Q. $I = \int_0^{\pi/4} \tan^2(x)\,dx = ?$

$$= \int_0^{\pi/4} (\sec^2 x - 1)\,dx$$

$$= \Big[\tan x - x\Big]_0^{\pi/4}$$

$$= \left(\tan\frac{\pi}{4} - \frac{\pi}{4}\right) - (\tan 0 - 0)$$

$$= 1 - \frac{\pi}{4}$$

Q. $I=\int_0^n \frac{1-\left(\frac{x}{n}\right)^n}{n-x}\,dx = ?$

$$= \int_0^n \frac{1-\left(\frac{x}{n}\right)^n}{n\left(1-\frac{x}{n}\right)}\,dx = \frac{1}{n}\int_0^1 \left(\frac{1-t^n}{1-t}\right)\cdot n\,dt$$

Put $\frac{x}{n}=t$

$x = nt$

$dx = n\,dt$

At $x=0,\ t=0$

At $x=n,\ t=1$

$$= \int_0^1 \left(\frac{1-t^n}{1-t}\right)dt$$

$$= \int_0^1 \left(1+t+t^2+\cdots+t^{n-1}\right)dt$$

$$= \left(t+\frac{t^2}{2}+\frac{t^3}{3}+\cdots+\frac{t^n}{n}\right)_0^1$$

$$= 1+\frac{1}{2}+\frac{1}{3}+\cdots+\frac{1}{n}$$

w.k.t for GP:

$$S_n = a+ar+ar^2+ar^3+\cdots+ar^{n-1} = \frac{a(1-r^n)}{1-r}$$

if $a=1, r=t$ then $S_n = 1+t+t^2+\cdots+t^{n-1} = \left(\frac{1-t^n}{1-t}\right)$

Properties of Definite Integral →

(1) $\int_a^b f(x)\,dx = \int_a^b f(t)\,dt$

(2) $\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$

(3) $\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx$

where $a < c < b$

(4) $\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$

(5) $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$

Property (6) →

$$\int_0^{2a} f(x)\,dx = \boxed{\int_0^a \{f(x)+f(2a-x)\}\,dx} = \begin{cases} 2\int_0^a f(x)\,dx, & \text{if } f(2a-x)=f(x) \\ 0, & \text{if } f(2a-x)=-f(x) \end{cases}$$

Property (7) →

$$\int_{-a}^{a} f(x)\,dx = \boxed{\int_0^a \{f(x)+f(-x)\}\,dx} = \begin{cases} 2\int_0^a f(x)\,dx, & \text{if } f(-x)=f(x) \text{ i.e. } f(x) \text{ is Even} \\ 0, & \text{if } f(-x)=-f(x) \text{ i.e. } f(x) \text{ is odd} \end{cases}$$

Prop (1) $I=\int_1^3 x\,dx=?=\left(\frac{x^2}{2}\right)_1^3=\frac{9-1}{2}=4$

$I=\int_1^3 t\,dt=?=\left(\frac{t^2}{2}\right)_1^3=\frac{9-1}{2}=4$

Prop (2) $I=\int_3^1 x\,dx=?=\left(\frac{x^2}{2}\right)_3^1=\frac{1-9}{2}$

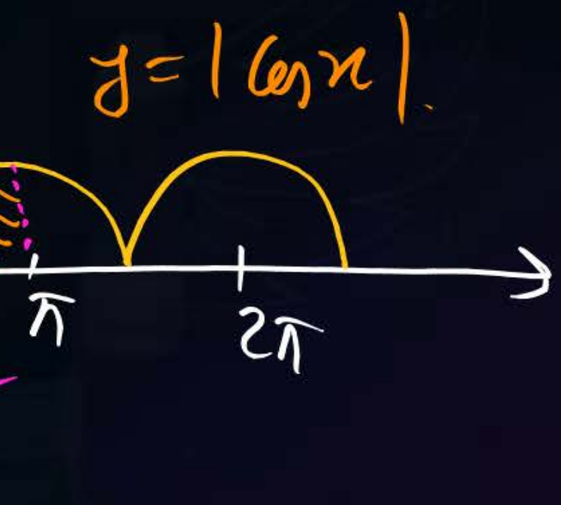
$=-4$

Prop (3) $I=\int_0^{\pi}|\cos x|\,dx=?$ (Ans = 2)

$=\int_0^{\pi/2}(+\cos x)\,dx+\int_{\pi/2}^{\pi}(-\cos x)\,dx$

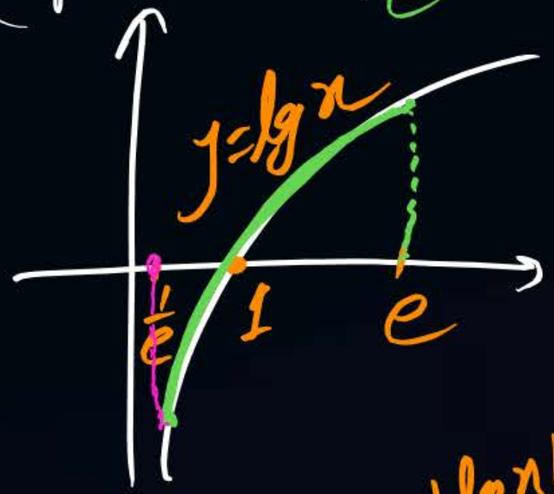
$=(\sin x)_0^{\pi/2}-(\sin x)_{\pi/2}^{\pi}$

$=(1-0)-(0-1)=2$



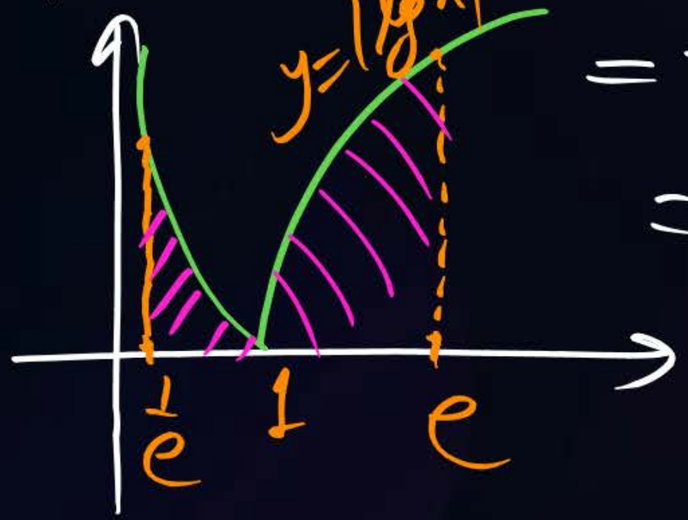
HW8 $I=\int_{1/e}^{e}|\log_e x|\,dx=?$ Ans: $2-\frac{2}{e}$

(Prop 3)



$$I=\int_{1/e}^{1}(-\lg x)\,dx+\int_{1}^{e}(+\lg x)\,dx$$

$$=-(x\lg x-x)\Big|_{1/e}^{1}+(x\lg x-x)\Big|_{1}^{e}$$

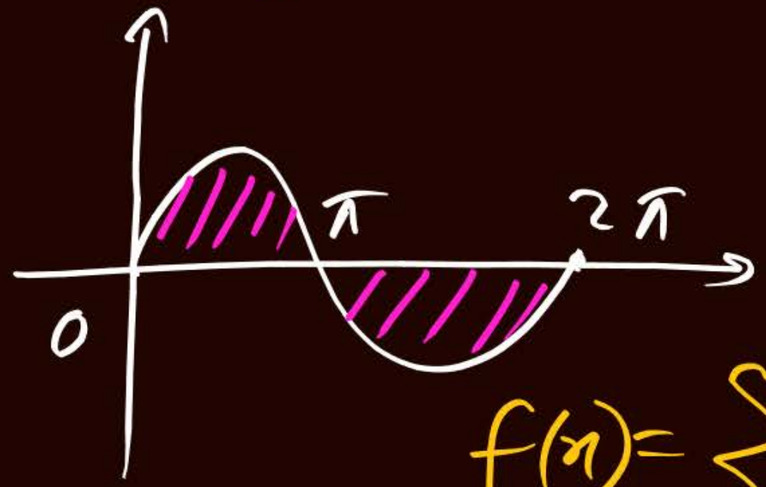


$$=-\left[(0-1)-\left(-\frac{1}{e}-\frac{1}{e}\right)\right]+\left[(e-e)-(0-1)\right]$$

$$=1-\frac{2}{e}+1=2-\frac{2}{e}$$

Q. $I = \int_0^{2\pi} \sin x\,dx = ? = (-\cos x)_0^{2\pi} = -(1-1) = 0$ ✓

Q. if $f(x) = \sin x$ then find area bounded by $f(x)$ b/n $x=0$ & $x=2\pi$



$$\text{Area} = \int_0^{2\pi} f(x)\,dx = \int_0^{\pi} (\sin x)\,dx + \int_{\pi}^{2\pi} (-\sin x)\,dx$$

$$f(x) = \begin{cases} \sin x, & 0 < x < \pi \\ -\sin x, & \pi < x < 2\pi \end{cases}$$

$$= (-\cos x)_0^{\pi} - (-\cos x)_{\pi}^{2\pi}$$

$$= -(-1-1) + (1-(-1))$$

$$= 2+2 = 4$$

Prop (4) :-

$$I=\int_{\pi/6}^{\pi/3}\frac{1}{1+\sqrt{\tan x}}\,dx=?=\int_{\pi/6}^{\pi/3}\frac{\sqrt{\cos x}}{\sqrt{\cos x}+\sqrt{\sin x}}\,dx \quad \text{— (1)}$$

$$a+b-x=\left(\frac{\pi}{6}+\frac{\pi}{3}-x\right)=\frac{\pi}{2}-x$$

$$I=\int_{\pi/6}^{\pi/3}\frac{\sqrt{\cos\left(\frac{\pi}{2}-x\right)}}{\sqrt{\sin\left(\frac{\pi}{2}-x\right)}+\sqrt{\cos\left(\frac{\pi}{2}-x\right)}}\,dx=\int_{\pi/6}^{\pi/3}\frac{\sqrt{\sin x}}{\sqrt{\cos x}+\sqrt{\sin x}}\,dx \quad \text{— (2)}$$

$$I+I=\int_{\pi/6}^{\pi/3}(1)\,dx=\frac{\pi}{3}-\frac{\pi}{6}=\frac{\pi}{6}\Rightarrow I=\frac{\pi}{12}$$ Ans

Prop (5) :→ $I = \int_0^{\pi/2} \log(\tan x)\, dx = ?$

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$ (c) 1 (d) 0

$$I = \int_0^{\pi/2} \log(\tan x)\, dx \quad \text{—(1)}$$

$$I = \int_0^{\pi/2} \log \tan\left(\frac{\pi}{2} - x\right) dx$$

$$I = \int_0^{\pi/2} \log(\cot x)\, dx \quad \text{—(2)}$$

$$I + I = \int_0^{\pi/2} (\log \tan x + \log \cot x)\, dx$$

$$2I = \int_0^{\pi/2} \log(1)\, dx$$

$$2I = 0$$

$$I = 0$$

Q. HW $I = \int_0^{\pi/4} \log(1+\tan x)\,dx = ?$

(a) $\pi/4$

(b) $\frac{\pi}{4}\log 2$

(c) 0

(d) $\frac{\pi}{8}\log 2$

THANK - YOU

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